

MAT 3150 — Homework 3

Problem 1 Walkthrough: Reverse Triangle Inequality

Goal. Prove that for any $x, y \in \mathbb{R}$,

$$||x| - |y|| \leq |x - y|.$$

Tools.

- Triangle inequality: $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
- For $c \geq 0$: $|a| \leq c \iff -c \leq a \leq c$.

Step 1 (worked)

Start from the identity

$$x = (x - y) + y.$$

Take absolute values and apply the triangle inequality:

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

Subtract $|y|$ from both sides:

$$|x| - |y| \leq |x - y|. \tag{*}$$

Step 2 (your turn)

Do the same thing, but start from

$$y = (y - x) + x.$$

Take absolute values and apply the triangle inequality:

$$|y| = |(y - x) + x| \leq |y - x| + |x|.$$

Subtract $|x|$ from both sides:

$$|y| - |x| \leq |y - x|. \tag{**}$$

Step 3 (combine)

Let $a = |x| - |y|$ and $c = |x - y|$. Note that $c \geq 0$.

Step 1 gives $a \leq c$. Inequality $(*)$ says $|x| - |y| \leq |x - y|$, which is literally $a \leq c$.

Step 2 gives $-a \leq c$. Rewrite both sides of $(\star\star)$:

- Left side: $|y| - |x| = -(|x| - |y|) = -a$.
- Right side: $y - x = -(x - y)$, and $|-t| = |t|$ for every real t , so $|y - x| = |x - y| = c$.

So $(\star\star)$ says $-a \leq c$. Multiplying by -1 flips the inequality: $a \geq -c$.

Together:

$$-c \leq a \leq c.$$

Which tool? The triangle inequality has already been used. The other tool,

$$|a| \leq c \iff -c \leq a \leq c \quad (c \geq 0),$$

has exactly $-c \leq a \leq c$ on its right-hand side, so it gives the left-hand side. (This is why we needed $c \geq 0$.)

Your turn: write $|a| \leq c$ with a and c replaced by what they stand for. That one line is the statement of Problem 1.

Checklist for the final write-up

- Both one-sided inequalities are stated and justified by the triangle inequality.
- You explicitly use $|y - x| = |x - y|$.
- You cite the fact $|a| \leq c \iff -c \leq a \leq c$ (with $c \geq 0$) to finish.